```
from google.colab import files
import cv2
import numpy as np
import matplotlib.pyplot as plt

uploaded = files.upload()
filename = list(uploaded.keys())[0]
image = cv2.imread(filename, 0)
print("Image uploaded successfully:", filename)

kernel = np.ones((7, 7), np.uint8)

opening = cv2.morphologyEx(
    image, cv2.MORPH_OPEN, kernel
)

plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.imshow(image, cmap="gray")
plt.title("Original Image with Noise")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(opening, cmap="gray")
plt.title("After Opening")
plt.axis("off")

plt.show()
```

Choose Files Screenshot… 075823.png

Screenshot 2026-10-10 075823.png(image/png) - 93416 bytes, last modified: 10/10/2026 - 100% done

```
Saving Screenshot 2026-10-10 075823.png to Screenshot 2026-10-10 075823 (8).png
Image uploaded successfully: Screenshot 2026-10-10 075823 (8).png
```



Getting started

The document you are reading is not a static web page, but an interactive environment called a **Colab notebook** that lets you write and execute code.

For example, here is a **code cell** with a short Python script that computes a value, stores it in a variable, and prints the result:

```
seconds_in_a_day = 24 * 60 * 60
seconds_in_a_day
```

```
86400
```

To execute the code in the above cell, select it with a click and then either press the play button to the left of the code, or use the keyboard shortcut "Command/Ctrl+Enter". To edit the code, just click the cell and start editing.